

VAPORIZATION THERMODYNAMICS AND KINETICS OF HEXAGONAL SILICON CARBIDE¹

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Mass spectrometer Knudsen effusion vaporization experiments over the temperature range 1835-2264 K confirm earlier mass spectrometric results that Si(g) is the predominant vapor species above the SiC-C two-phase region. Si₂C(g) and SiC₂(g) are found to be the next most important vapor species. Total vapor pressures using a gold calibrated mass spectrometer are measured to be a factor of 10-15 higher than values previously reported between 1800-2200 K. Partial pressures of Si(g), SiC₂(g), and Si₂C(g) lead to third-law enthalpies of formation for SiC₂(g) and Si₂C(g) which are in accord with previously recommended values.

Langmuir vaporization of SiC(hexagonal) (0001) single crystal faces in the mass spectrometer show ion intensities of Si⁺ and SiC₂⁺ that are time-independent after an initial induction period during which the intensities increase by a factor of 1.5 before attaining a steady-state value. The mass spectrometer studies between 2200-2500 K give activation enthalpies of $\Delta H^*(2350) = 141.6 \pm 2.7 \text{ kcal mol}^{-1}$ for Si(g) and $\Delta H^*(2350) = 166.2 \pm 3.5 \text{ kcal mol}^{-1}$ for SiC₂(g) vaporization from the (0001) crystal face of hexagonal SiC.

Constant temperature total mass-loss Langmuir vaporization experiments using SiC (0001) crystal faces show a time-dependent rate of mass-loss. It is believed that this time-dependent behavior is caused by a decrease in the SiC(hexagonal) single crystal surface area and is not due to impedance of silicon vaporization by the graphite layer formed on the crystal surface during decomposition.

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1. Introduction

Low atomic number ceramic solids have numerous potential uses in controlled thermonuclear reactor engineering applications [1-3]². Ceramic materials being considered for such applications include SiC, Be₂C, B₄C, BN, Si₃N₄, Al₂O₃, and BeO. These materials are also of interest from a fundamental point of view as many of them vaporize by incongruent decomposition, often yield complex and unique vapor molecules at high temperatures, and are believed to exhibit vacuum vaporization rates lower than the equilibrium values.

The present paper reports the results of investigations on the vaporization thermodynamics of hexagonal silicon carbide using high temperature quadrupole mass spectrometry. Preliminary results of investigations using mass spectrometry to study the vaporization of hexagonal SiC single crystals are also reported. The vaporization thermodynamics and kinetics of hexagonal SiC were investigated because:

- (1) of the potential application of SiC as a first wall protective barrier in fusion reactors,
- (2) discrepancies exist in the literature regarding the vaporization thermodynamics of SiC and the composition of the equilibrium vapor,
- (3) the vaporization kinetics of SiC have not been previously established, and
- (4) single crystal wafers of hexagonal SiC can be readily obtained.

2. Summary of Previous Work on SiC

Silicon carbide, SiC, is believed to be the only compound formed in the SiC system [4,5]. SiC exists as two phases: an alpha or hexagonal phase and a beta or cubic phase. Cubic SiC is believed to be the thermodynamically stable phase at low temperatures. The cubic to hexagonal phase transition temperature is reported to lie between 2000-2300 K [6]. Thermodynamic stabilities of the two SiC phases are believed to be similar, their free energies of formation differing by about 0.5 kcal mol⁻¹ at 298 K. Hexagonal SiC forms numerous polytypic modifications which arise from different stacking sequences of the hexagonal SiC layers [7]. Differences in the thermodynamic stabilities of these polytypes have not been determined.

Quantitative data regarding the composition range over which SiC exists have not been reported. It is generally assumed that SiC exists as a line compound [5].

SiC vaporizes incongruently to form solid graphite as a reaction product. Vapor pressures in both the SiC-C and Si-SiC systems have been investigated since as early as 1926. Ruff and Konschak [8] and Ruff and Grieger [9] measured the vapor pressure of the SiC-C system between 2673-2990 K and 2565-2951 K, respectively. Assuming the vapor to consist of Si(g) and SiC(g), these authors were able to calculate Si(g) partial pressures for the SiC-C system.

²Figures in brackets indicate the literature references at the end of this paper.

Drowart, DeMaria, and Inghram [10] used mass spectrometry to establish the vapor composition in the SiC-C system between 2149-2316 K. They determined that Si(g) is the predominant vapor species with SiC₂(g) and Si₂C(g) being the next most predominant vapor species over their experimental temperature range. They also found Si₂(g), SiC(g), Si₂C₂(g), Si₃(g), Si₂C₃(g), and Si₃C(g) to be minor species in the equilibrium vapor.

While investigating the spectroscopy of SiC molecules by matrix isolation techniques, Weltner and McLeod [11] found a relatively strong Si₂C₃ spectrum. As they could not determine whether the strong spectrum was due to a high transition probability of the Si₂C₃ bands in the visible region of the spectrum or to a concentration of Si₂C₃ in the SiC-C vapor higher than that suggested by the mass spectrometry results of Drowart, et al., Weltner and McLeod suggested that the importance of this molecule in the high temperature equilibrium vapor of SiC-C be re-examined.

Knippenberg [5] measured the total vapor pressure of hexagonal SiC in a graphite crucible and determined the silicon content of the vapor at the normal sublimation temperature (3030 K from Knippenberg's work). On the basis of his analysis, Knippenberg concluded that the equilibrium gas phase composition of the SiC-C system is always rich in silicon in contrast to the results of the mass spectrometer studies of Drowart, et al., which indicate the Si/C ratios in the gas phase to be less than unity.

Drowart and DeMaria [12] measured the vapor pressure of the Si-SiC system using mass spectrometry and found Si(g) to be the major vapor species with Si₂C(g) and SiC₂(g) being the next most important vapor species below 2632 K. Enthalpies of formation of Si₂C(g) and SiC₂(g) calculated by a third-law analysis of the vapor pressure data are consistent with the corresponding third-law values from Drowart, et al.'s mass spectrometer work on SiC-C. Total vapor pressures extrapolated to the temperature range over which Ruff and co-workers [8,9] measured total vapor pressures of the Si-SiC system are in poor agreement with Ruff's results. Drowart and DeMaria attributed this discrepancy to the fact that silicon diffused through the crucible in Ruff's experiments causing their vapor pressures to be inordinately high.

Grieverson and Alcock [13] used a mass-loss effusion technique to measure the total vapor pressure of both hexagonal and cubic SiC. The measured total vapor pressures for each SiC phase were slightly higher than those reported by Drowart, et al. At temperatures above 1973 K, Grieverson and Alcock found that their vapor pressures for hexagonal SiC were higher than values calculated by extrapolating their low temperature data. This behavior may indicate that the low temperature vapor pressures are too low because of kinetic problems (i.e., a low vaporization coefficient).

Davis, Anthrop, and Searcy [14] measured the vapor pressures of cubic SiC by a mass-loss effusion technique using Drowart, et al.'s mass spectrometer results to calculate Si(g) partial pressures from their observed mass losses. They obtained total vapor pressures in agreement with the total vapor pressures measured by Grieverson and Alcock. Davis, et al. also report that they found no systematic change in the vapor pressure as a function of cell orifice area and concluded that the vaporization coefficient of Si(g) from cubic SiC was close to unity. They did, however, observe a time-dependent decrease in the

vaporization rate which they attributed to impedance of Si(g) vaporization by a graphite microlayer formed on the SiC particles during vaporization.

Mell and Brodsky [15] obtained the mass spectrum of silicon evaporating from a carbon-lined copper hearth under high vacuum (The temperature at which the mass spectrum was taken was not reported.). They observed peaks for most of the known Si-C and Si molecules but did not observe a peak attributable to SiC₂(g).

Mass-loss effusion studies of SiC-C by Voronin, Makarova, and Yudin [16] give vapor pressures which are a factor of 20-30 below those of the previously discussed investigations. These results indicate a low vaporization coefficient in contradiction to the results of Davis, et al. Ghoshtagore [17] performed Langmuir vaporization experiments using hexagonal SiC single crystal wafers and reported vaporization rates a factor of 10⁴ lower than those measured under equilibrium conditions.

3. Experimental Section

3.1 Sample characterization

Knudsen effusion experiments are performed using 200 mesh SiC particles, 200 mesh SiC particles crushed to 325 mesh powder, or 325 mesh SiC powder mixed with 325 mesh H-451 reactor grade graphite powder. X-ray diffraction shows the SiC to be the hexagonal phase, most probably a mixture of polytypes. Cubic SiC may also be present as x-ray diffraction lines for the cubic phase are not clearly discernible from those for the hexagonal phase. The x-ray diffraction patterns also show that significant amounts of impurities are not present in the SiC samples.

Langmuir vaporization experiments are performed using hexagonal SiC single crystal wafers purchased from the Materials Research Corporation, Orangeburg, N.Y. The crystals are carbon arc grown, 6-8 mm in cross section and 1 mm thick with the c-axis oriented perpendicular to the large face. X-ray diffractometer scans confirm the crystals to be hexagonal SiC.

3.2 Experimental apparatus

The mass spectrometer used for both the Knudsen effusion and Langmuir vaporization experiments is shown in figures 1 and 2. The mass spectrometer consists of an Extra-nuclear Laboratories field separation (ELFS) quadrupole mass filter. The quadrupole rods are 0.95 cm diameter by 20 cm long giving a mass range of 1400 amu with a resolution of approximately 1000 at m/e = 1000.

The electron impact ionizer is of cross-beam design; that is, the molecular beam from the high temperature source enters the ionizer region in a direction normal to the geometrical axis of the quadrupole rods and normal to the direction of ion extraction into the mass filter.

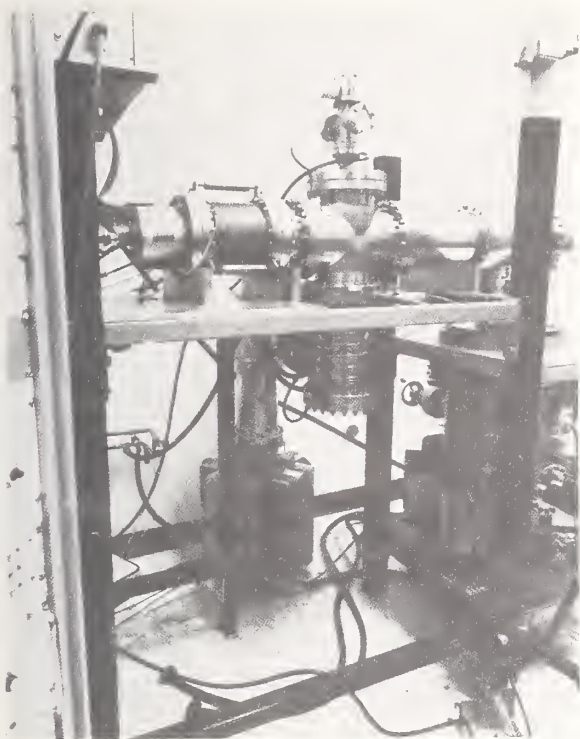


Figure 1. High temperature mass spectrometer. View 1.

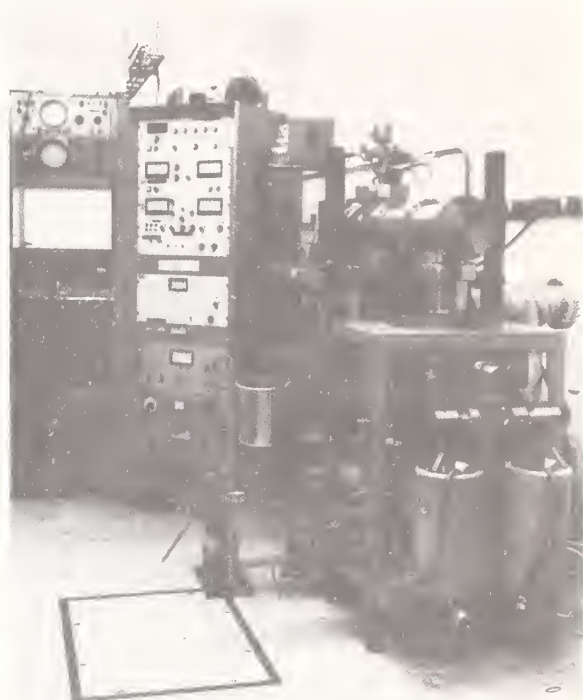


Figure 2. High temperature mass spectrometer. View 2.

Ion detection is achieved by a 21-stage copper-beryllium "venetian blind" electron multiplier mounted paraxial to and offset from the quadrupole transmission axis. Multiplier gains are determined in the usual manner using a Faraday plate detector.

Multiplied ion currents pass through a preamplifier, through a 10^7 or 10^9 ohm resistor, and into a dc electrometer. Output from the electrometer is displayed on an oscilloscope, X-Y recorder, or a strip chart recorder.

The electron multiplier-mass filter-ionizer assembly is mounted as a single unit on an 8-inch o.d. conflat flange through which electrical connections to the quadrupole power supply, ionizer control, and electrometer are made. The assembly is mounted horizontally on a 5.5-inch i.d. stainless steel bellows which can be adjusted to permit easy alignment of the ion source entrance hole with the molecular beam from the high temperature Knudsen cell region (see fig. 1).

The vacuum system is constructed entirely of 6-inch o.d. stainless steel tubing. Conflat flanges with copper gaskets are used in all vacuum connections. A Varian FC-12E vacuum station is used to pump on the ion source region. The station consists of 200 l s^{-1} of Triode VacIon pumping, a titanium sublimation pump (TSP), and a liquid nitrogen cryo-array. A poppet valve is located above the VacIon pumps and TSP to permit breaking of vacuum without turning off the pumps. Pumping of the Knudsen cell region is achieved

by a 400 l s^{-1} Thermionics Laboratory VacIon pump. Rough pumping of the system is achieved by three liquid nitrogen-cooled VacSorb pumps. These pumps are isolated from the main vacuum system by 1.5-inch polyimide-sealed ultra-high vacuum valves (see fig. 2).

Typical background pressures of 5×10^{-9} Torr are readily achieved in both the ion source and Knudsen cell regions after about six hours pumping from 10 microns pressure. Background pressures on the order of 1×10^{-7} Torr or better in the ion source region and 1×10^{-6} Torr or better in the Knudsen cell region are routinely achieved during operation with a graphite Knudsen cell at temperatures as high as 2300 K.

The Knudsen cell region is separated from the ion source region of the mass spectrometer by a water-cooled plate with a 0.25-inch diameter hole through its center. A stainless steel shutter blade with a 0.025-inch wide slit etched into it rests on top of the water-cooled plate and is moved across the path of the molecular beam from the Knudsen cell region by a linear-motion vacuum feedthrough.

A graphite Knudsen effusion cell or SiC single crystal situated in a graphite holder is heated by a tungsten spiral resistance heater mounted on an 8-inch o.d. conflat flange (fig. 3). A Sorenson high current, low voltage dc power supply is used to supply power to the tungsten heater. Water-cooled copper power feedthroughs for the tungsten heater are brought into the vacuum system through the 8-inch o.d. flange. A concentric arrangement of six tantalum and two tungsten shield cans, with molecular beam exit holes drilled through the center of their tops, is situated around the tungsten heater. Seven tantalum heat shields, with pyrometer sight holes drilled through their centers, are situated directly below the heater. The entire furnace arrangement fits into a 6-inch o.d. stainless steel housing which has 0.25-inch o.d. copper tubing brazed to the outside wall for water cooling.

The temperature of a Knudsen cell is measured using a calibrated Pyro optical pyrometer by sighting through an optical window onto a blackbody hole drilled into the bottom of the Knudsen cell. Single crystal temperatures are measured by sighting the pyrometer through a hole drilled in the bottom of a cylindrical graphite holder onto the underside of the single crystal. Observed temperatures are then corrected for optical window and prism effects.

Knudsen effusion cells used in the equilibrium vapor pressure measurements are machined from graphite and are of two types. The first type consists of a base with a lid, has a 1.3 cm outside diameter and is approximately 1.8 cm long. The cell lid contains a knife-edge circular orifice with an area of $3.0 \times 10^{-3} \text{ cm}^2$ and a Clausing factor of 0.502. The second type of Knudsen effusion cell consists of a single piece of graphite 1.8 cm in length and 1.3 cm in outside diameter. A 0.48 cm long graphite plug with a cylindrical hole drilled through its center provides the orifice. The orifice area is 0.027 cm^2 with a Clausing factor of 0.385.

Total mass-loss Langmuir vaporization experiments are performed with a SiC single crystal wafer sitting in a graphite holder. The crystal and holder are seated in the center of a water-cooled eddy-current concentrator on a Veeco vacuum pumping station. The current concentrator is coupled to induction coils from a Lepel 20 kW induction heating

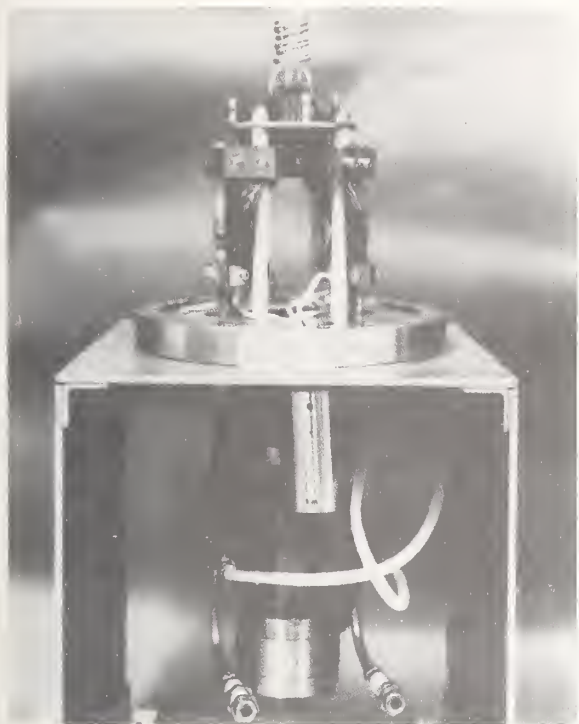


Figure 3. Tungsten spiral resistance heater on 8-inch conflat flange.

unit. The crystal surface temperature is measured by sighting a Pyro optical pyrometer through a quartz window located at the top of the concentrator onto the crystal surface. The mass change of the crystal after heating for a given time period is determined by removing the crystal from the concentrator and weighing it on a Mettler pan micro-balance.

3.3 Data treatment

Ion intensities from the SiC Knudsen effusion studies are converted to absolute partial pressures for each vapor species assuming the ion transmission efficiency of the mass filter is independent of m/e^3 , using the results of a separate Knudsen effusion vaporization of gold from a graphite crucible, and using the relation

$$P_u = C_{Au} I_u T \frac{(\sigma \gamma f i)_{Au}}{(\sigma \gamma f i)_u} \quad (1)$$

where P_u is the pressure of molecular species u with measured ion intensity I_u ; C_{Au} is the instrument sensitivity; T is the thermodynamic temperature; γ is the electron multiplier

³Editor's note: This is not necessarily a good assumption, e.g., see Section 5.1 and the chapter by Bonnell and Hastie in this volume.

gain; f is the isotopic abundance; i is an emission current correction factor; and σ is the calculated ionization cross section at the working electron energy. The instrument sensitivity, C_{Au} , is obtained from the gold vaporization experiment and the known vapor pressure of gold [18].

Vapor pressures obtained using eq. (1) are fitted to the relation

$$R \ln(P/\text{atm}) = -\Delta H^\circ/T + \Delta S^\circ \quad (2)$$

by a least-squares calculation. Second-law values of $\Delta H^\circ(298)$ and $\Delta S^\circ(298)$ are calculated by a Σ' evaluation of the vapor pressures [19]. Third-law values of $\Delta H^\circ(298)$ are obtained in the usual manner [20].

Time-independent ion intensities from the single crystal Langmuir vaporization studies are fitted to the relation

$$R \ln(IT) = -\Delta H^*/T + \text{Constant} \quad (3)$$

by a least-squares calculation. The quantity ΔH^* is the activation enthalpy change for the vaporization process. The similar form of eqs. (2) and (3) permits ΔH^* and ΔH° to be compared to yield an enthalpy barrier, $\Delta H^* - \Delta H^\circ$, for the kinetically slow step(s) in the vaporization process.

4. Experimental Results

4.1 Knudsen effusion mass spectrometer measurements on the SiC-C system

The most important vapor molecule in the SiC-C system over the experimental temperature range of 1835-2264 K is found to be Si(g) with Si₂C(g) and SiC₂(g) being the next most important molecules. Only trace amounts of shutterable Si₂(g), Si₂C₂(g), Si₃(g), Si₃C(g), and C₃(g) are detected to be present at 2275 K. Mass spectrum peaks attributable to ions of other silicon or silicon-carbon molecules are not detected. These results are in accord with the mass spectrometer results of Drowart, et al. [10]. X-ray analysis of the solid product remaining after complete vaporization of silicon shows it to be graphite.

Temperature coefficient plots for Si⁺ (m/e = 28), SiC₂⁺ (m/e = 52), and Si₂C⁺ (m/e = 68) are shown in figure 4 for a typical mass spectrometer experiment using a graphite effusion cell with a $3 \times 10^{-3} \text{ cm}^2$ orifice area and 325 mesh SiC powder. For this particular experiment, electron energies of 13 eV are used for the Si⁺ intensity measurements and 15 eV for the SiC₂⁺ and Si₂C⁺ measurements. An emission current of 2.0 mA is used for all the measurements.

Figure 4 shows attenuation of the Si⁺, Si₂C⁺, and SiC₂⁺ intensities at high temperatures. This behavior is believed to be caused by intramolecular scattering in the molecular beam. Ion intensity attenuation caused by intramolecular scattering as predicted by a model developed by Meschi [21,22] is indicated in figure 4 by the dashed lines. The total

vapor pressure inside the SiC-containing graphite Knudsen cell at which attenuation is first observed is in good agreement with that predicted by Meschi's model. However, agreement between the magnitude of the observed attenuation and that predicted by Meschi's model is only qualitative. This discrepancy may be caused by incorrect assumptions contained in the model, incorrect assumptions concerning collision cross-sections for Si-C molecules used in the calculations, or by an added contribution due to scattering of molecules in the molecular beam by background gases.

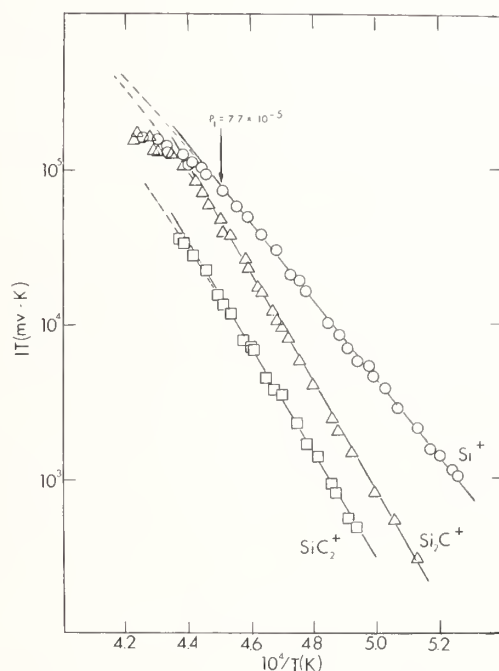


Figure 4. Temperature coefficient plot of Si^+ , SiC_2^+ , and Si_2C^+ for a typical mass spectrometer Knudsen effusion experiment using a graphite effusion cell with $3 \times 10^{-3} \text{ cm}^2$ orifice area. Dashed lines indicate expected intensity attenuation calculated using Meschi's model for intramolecular scattering [21].

To assure that the observed ion intensity attenuation is not caused by some anomalous behavior of the mass spectrometer or by vaporization kinetic problems (e.g., solid state diffusion of silicon is determining the vaporization rate at high temperature), vaporization of molten gold from a graphite Knudsen cell was studied at temperatures where the gold vapor pressure is on the order of 10^{-3} - 10^{-4} atmospheres. A temperature coefficient plot for this experiment is shown in figure 5 (data are taken using 20 eV electron energy,

3.0 mA emission current). Attenuation of the Au^+ ($m/e = 197$) intensities starts at gold vapor pressures predicted from Meschi's model (about 5×10^{-4} atmospheres). The magnitude of the attenuation is also comparable to that predicted by the model.

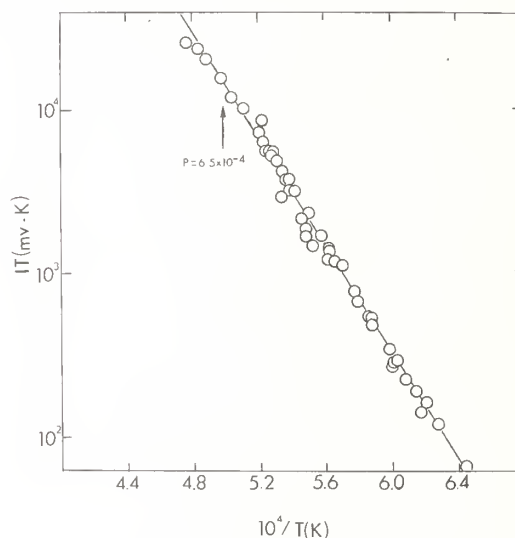
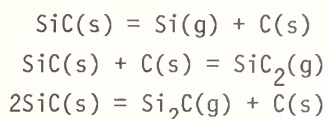


Figure 5. Temperature coefficient plot of Au^+ data from mass spectrometer Knudsen effusion experiment using a graphite effusion cell with $3 \times 10^{-3} \text{ cm}^2$ orifice area.

Measured partial pressures of Si(g) , $\text{SiC}_2(\text{g})$, and $\text{Si}_2\text{C(g)}$ are found to depend on both the SiC particle size and the size of the Knudsen cell orifice. SiC powder ground to 325 mesh (44 micron) and vaporized from a Knudsen cell with a $3 \times 10^{-3} \text{ cm}^2$ orifice area gives the highest measured vapor pressures. A coarsely crushed hexagonal SiC single crystal vaporized from the crucible with a $3 \times 10^{-3} \text{ cm}^2$ orifice area gives partial pressures 90 percent below vapor pressures obtained with the 325 mesh samples. Vapor pressures measured using 200 mesh SiC powder and a crucible with a 0.027 cm^2 orifice area are 65 percent lower than those obtained using the 35 mesh powder in an effusion cell with a $3 \times 10^{-3} \text{ cm}^2$ orifice area. Samples of 200 mesh SiC powder vaporized in the effusion cell with a $3 \times 10^{-3} \text{ cm}^2$ orifice area give measured partial pressures the same as obtained with the 325 mesh samples.

Because of the orifice and particle size effects, thermodynamic quantities for SiC vaporization are calculated using data from those experiments with 200 and 325 mesh SiC powder and a $3 \times 10^{-3} \text{ cm}^2$ orifice area. Second- and third-law enthalpies of vaporization at 298 K for the reactions



are summarized in table 1. These values are obtained as described in Section 3.3 and using JANAF [6] free energy functions. Vapor pressures computed using the present third-law enthalpy changes and JANAF free energy functions are given in table 2.

4.2 Mass spectrometer and total mass-loss Langmuir vaporization experiments

Two mass spectrometer experiments, performed over the temperature range 2200-2500 K, investigated the Langmuir vaporization from the (0001) basal plane of two separate SiC single crystals. Shutterable ions observed are Si^+ , SiC_2^+ , and Si_2C^+ . Ion intensities are found to be independent of time after an initial induction period during which the intensity of Si^+ (and presumably SiC_2^+ and Si_2C^+) increases with time. The steady-state Si^+ intensity is a factor of 1.5 higher than that measured during the initial stages of heating. At 2400 K, measured steady-state ion intensities are a factor of 80-100 lower than intensities extrapolated from those measured in the Knudsen effusion experiments.

The measured temperature-dependence of Si^+ and SiC_2^+ intensities yield activation enthalpies of $\Delta H^*(2350) = 141.6 \pm 2.7 \text{ kcal mol}^{-1}$ for Si(g) and $\Delta H^*(2350) = 166.2 \pm 3.5 \text{ kcal mol}^{-1}$ for $\text{SiC}_2\text{(g)}$ sublimation from the (0001) plane of hexagonal SiC. Comparison of these activation enthalpies with the corresponding equilibrium enthalpies obtained in the present work, $\Delta H^0(2350) = 111.3 \text{ kcal mol}^{-1}$ and $\Delta H^0(2350) = 143.7 \text{ kcal mol}^{-1}$, respectively, gives enthalpy barriers of $30.3 \text{ kcal mol}^{-1}$ and $22.5 \text{ kcal mol}^{-1}$ for Si(g) and $\text{SiC}_2\text{(g)}$ sublimation.

Results of a total mass-loss Langmuir vaporization study of the SiC (0001) face are shown for one experiment conducted at 2498 K in figure 6. The results show that the total mass-loss is time-dependent during the entire course of vaporization (i.e., from initial heat-up until all the silicon is vaporized from the crystal). Figure 7 shows the mass-loss data of figure 6 plotted against $t^{1/2}$ indicating a parabolic time-dependence. Figure 8 shows the time-dependence of the apparent flux vaporizing from the (0001) crystal face calculated from the data shown in figure 6. It should be noted that the flux measured initially ($0.31 \text{ mg min}^{-1} \text{ cm}^2$) is a factor of about 10^4 below the equilibrium flux of Si(g) at 2498 K, and that during the 1300 minute vaporization, the apparent flux drops by a factor of four. The flux then immediately drops to zero at which time it is believed that the silicon is completely depleted from the crystal.

Table 1. Comparison of vaporization enthalpies for the SiC(hex)-C system.

Reaction	$\Delta H^\circ(298)$, second-law (kcal mol ⁻¹)		$\Delta H^\circ(298)$, third-law (kcal mol ⁻¹)	
	<u>This work</u>	<u>Reference [10]^a</u>	<u>This work</u>	<u>Reference [10]^a</u>
SiC(s) = Si(g) + C(s)	119.2±2.1 ^b	136.9±0.7 ^b	115.2±0.8	124.6±0.3
2SiC(s) = Si ₂ C(g) + C(s)	162.4±2.1	169.6±7.5	145.3±1.2	161.3±0.5
SiC(s) + C(s) = SiC ₂ (g)	169.0±6.4	168.1±9.4	152.1±1.6	163.6±0.6

^aRecalculated from partial pressures reported in reference [10] using JANAF free energy functions.

^bAll uncertainties are quoted as one standard deviation.

Table 2. Comparison of calculated equilibrium pressures in the SiC(hex)-C system.

Molecule	T(K)	P(atm)	
		<u>This work^a</u>	<u>Reference [10]^a</u>
Si	1800	1.2×10^{-6}	8.7×10^{-8}
	2000	2.9×10^{-5}	2.7×10^{-6}
	2200	3.8×10^{-4}	4.4×10^{-5}
SiC ₂	1800	3.6×10^{-8}	1.5×10^{-9}
	2000	2.3×10^{-6}	1.3×10^{-7}
	2200	6.6×10^{-5}	4.7×10^{-6}
Si ₂ C	1800	1.2×10^{-7}	1.3×10^{-9}
	2000	5.7×10^{-6}	1.0×10^{-7}
	2200	1.4×10^{-4}	3.5×10^{-6}
P _{total}	1800	1.4×10^{-6}	9.0×10^{-8}
	2000	3.7×10^{-5}	2.9×10^{-6}
	2200	5.9×10^{-4}	5.2×10^{-5}

^aCalculated using third-law values of $\Delta H^\circ(298)$ given in table 1 and JANAF free energy functions.

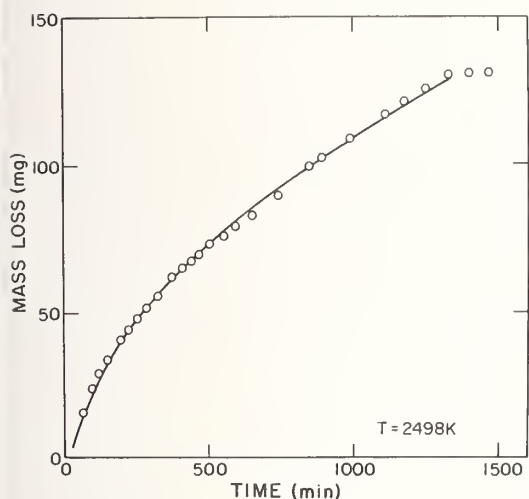


Figure 6. Time-dependence of total mass-loss from a freely vaporizing hexagonal SiC (001) single crystal face at 2498 K.

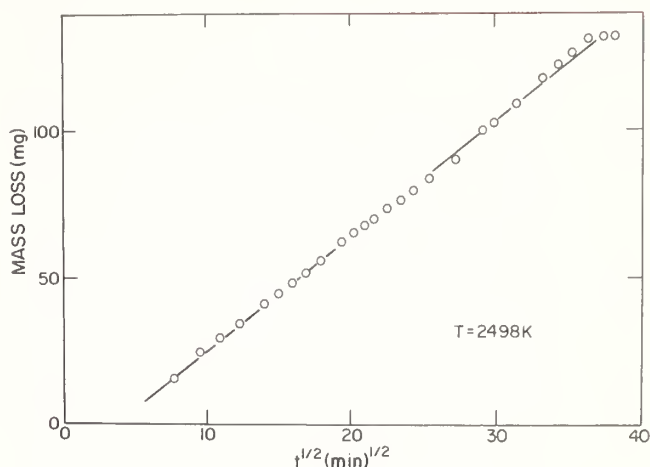


Figure 7. Total mass-loss data of figure 6 showing parabolic time-dependence.

The residue formed on the SiC single crystal substrate was investigated using x-ray diffraction and scanning electron microscopy (SEM). X-ray diffraction scans show the residue to be graphite. Broadening of the 002 and 004 graphite diffraction peaks indicates the graphite surface to be composed of small graphite crystallites. Scanning electron microscopy shows no pores to be present at magnification of up to 50,000X.

5. Discussion

5.1 Knudsen effusion mass spectrometer experiments

Over the temperature range 1800-2200 K equilibrium partial pressures of Si(g), SiC₂(g), and Si₂C(g) above SiC-C measured in the current work using a gold calibrated mass spectrometer are higher than pressures reported by Drowart, et al. [10] by factors of 10-15, 15-25, and 40-90, respectively. Accordingly, total vapor pressures of the present work are a factor of 10-15 higher than those of Drowart, et al. [10]. Present total vapor pressures extrapolated to 2800-2900 K are in excellent agreement with total pressures measured at these temperatures by Ruff and Konschak [8] and Ruff and Grieger [9].

Table 1 shows excellent agreement between present second- and third-law values of $\Delta H^\circ(298)$ for reaction (1). Agreement between second- and third-law enthalpies for reactions (2) and (3) is not good, the second-law enthalpies being 17 kcal mol⁻¹ lower than the corresponding third-law values. Note, however, that the present second-law enthalpies for reactions (2) and (3) are in good agreement with the corresponding third-law enthalpies reported by Drowart, et al.

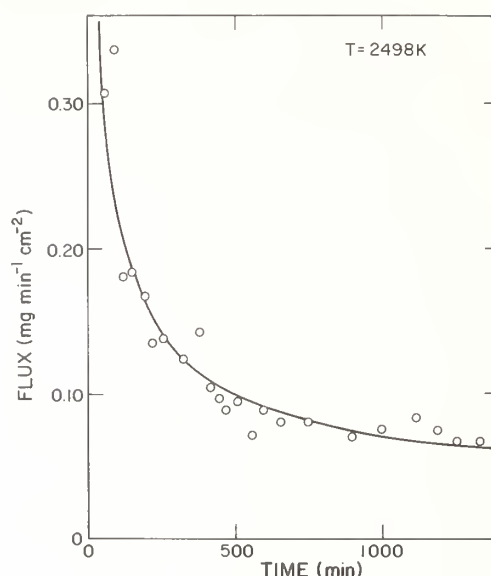


Figure 8. Time-dependence of total flux vaporizing from the (0001) SiC single crystal face at 2498 K. Calculated from data in figure 6.

The partial vapor pressures of Si(g) , $\text{SiC}_2\text{(g)}$, and $\text{Si}_2\text{C(g)}$ reported in table 2 for the present work are believed to be too high and the corresponding third-law enthalpy changes are believed to be too low for the following reasons. Ion intensities from the mass spectrometer studies were converted to absolute partial pressures using the results of a separate gold calibration experiment assuming the ion transmission efficiency of the quadrupole mass filter to be independent of m/e . However, because of the wide mass range between $\text{Au}^+(\text{m/e}=197)$ and $\text{Si}^+(\text{m/e}=28)$, the transmission efficiency of the mass filter may not be independent of m/e thus leading to erroneous vapor pressures in the present work. Vapor pressures of the present work compared to those reported by Drowart, et al. suggest that the transmission Au^+ in our mass filter is about ten percent relative to the transmission of Si^+ (the transmission of Au^+ also appears to be about ten percent relative to the transmission of SiC_2^+ and Si_2C^+ as well). Other indications of higher transmission efficiency for low mass ions than for high mass ions in the mass filter are:

1. The vapor pressure of $\text{C}_3\text{(g)}$ above graphite calculated from the present measure C_3^+ intensity at 2275 K and the gold calibration results is a factor of thirty higher than the vapor pressure calculated from thermodynamic data presented in the JANAF tables.

2. Mass spectrometer Knudsen effusion experiments on $\text{Al}_4\text{C}_3(\text{s})$ and $\text{Al}_4\text{SiC}_4(\text{s})$, both of which vaporize incongruently to give $\text{Al}(\text{g})$, also indicate the transmission of Au^+ to be approximately ten percent relative to $\text{Al}^+(\text{m/e}=27)$.
3. The enthalpy of formation of SiC (hexagonal) at 298 K computed from the present third-law enthalpy change given in table 1 for $\text{Si}(\text{g})$ vaporization from SiC and the enthalpy of formation for $\text{Si}(\text{g})$ given in the JANAF tables, $107.7 \text{ kcal mol}^{-1}$ at 298 K, is $-7.5 \text{ kcal mol}^{-1}$. This value is $9.4 \text{ kcal mol}^{-1}$ more positive than that recommended by JANAF based on the enthalpy of reaction for $\text{SiC}(\text{hex}) + 4\text{F}_2(\text{g}) = \text{SiF}_4(\text{g}) + \text{CF}_4(\text{g})$ measured by fluorine bomb calorimetry.

Enthalpies of formation for $\text{Si}_2\text{C}(\text{g})$ and $\text{SiC}_2(\text{g})$ are computed to be $130.3 \text{ kcal mol}^{-1}$ and $144.6 \text{ kcal mol}^{-1}$, respectively, using the present third-law enthalpy changes for reactions (1) through (3) given in table 1. Third-law values of ΔH° (298 K) for reactions (2) and (3) derived using a mass spectrometer calibration constant obtained by normalizing the present Si^+ intensity at 2000 K to the partial pressure of $\text{Si}(\text{g})$ above $\text{SiC}(\text{hex})$ reported by Drowart, et al., give $\Delta H_f^\circ(298 \text{ K}) = 121.1 \text{ kcal mol}^{-1}$ and $144.8 \text{ kcal mol}^{-1}$ respectively for $\text{Si}_2\text{C}(\text{g})$ and $\text{SiC}_2(\text{g})$. These are in excellent agreement with values recommended by JANAF based on the mass spectrometer results of Drowart, et al. [10,12], $(128 \pm 6) \text{ kcal mol}^{-1}$ and $(147 \pm 7) \text{ kcal mol}^{-1}$, respectively.

5.2 Langmuir vaporization experiments

Activation enthalpy barriers of $30.3 \text{ kcal mol}^{-1}$ for $\text{Si}(\text{g})$ and $22.5 \text{ kcal mol}^{-1}$ for $\text{SiC}_2(\text{g})$ vaporization from the $\text{SiC}(\text{hexagonal})$ basal plane lead to vaporization coefficients of 1.5×10^{-3} and 8.1×10^{-3} , respectively, at 2350 K if it is assumed there are no corresponding entropy barriers for the kinetically-slow step in the vaporization process (i.e., $\Delta S^\circ = \Delta S^*$).

The time-independent Si^+ and SiC_2^+ intensities observed in the mass spectrometer Langmuir single crystal experiments show that the porous graphite layer formed as a result of SiC decomposition has little or no effect on the transport rate of $\text{Si}(\text{g})$ and $\text{SiC}_2(\text{g})$ from the crystal surface. The parabolic time-dependence of the total mass-loss observed in the mass-loss Langmuir experiments is believed to be due to a decrease in the surface area of the single crystal as decomposition proceeds and not due to a diffusion-controlled rate-limiting step in the vaporization process. A time-dependent decrease in Si^+ , SiC_2^+ , and Si_2C^+ intensities due to a decrease in the single crystal surface area is not expected to be observed in the Langmuir mass spectrometer experiments as only a small portion of the vapor flux from the center of the SiC crystal surface is sampled by the mass spectrometer.

A simple model used to calculate the mass-loss of a cube with a surface area decreasing with time gives mass-losses in qualitative agreement with those observed experimentally. The data shown in figure 6 best fit the model at short times ($t < 500$ minutes). At longer times, the model predicts a too-slow decrease in the surface area with time and, hence, too fast a loss of material in comparison with the experimental data.

Details concerning the mechanism for the decomposition of SiC cannot be presented at this time. However, on the basis of a model developed by Searcy for endothermic decomposition reactions which form porous solid reaction products, the preliminary vaporization data and SEM results reported here indicate that either a chemically activated mechanism of surface diffusion or a step involving the gaseous reaction products is rate-limiting [24,25].

We wish to thank Alan Searcy for helpful discussions concerning the kinetics of decomposition reactions and for informing us of Meschi's model for intramolecular scattering; MaryAnn David for performing some of the mass spectrometer Knudsen effusion measurements; and Terry Wallace for his support and interest in this project. We wish also to thank the Los Alamos Scientific Laboratory for providing funds under its New Research Initiatives Program.

References

- [1] Rovner, L. H. and Hopkins, G. R., Nuclear Tech. 29, 274 (1976).
- [2] Hopkins, G. R., Low Atomic Number Materials for Fusion. An Assessment Study, General Atomic Company Report GA-A13306 (September 15, 1974).
- [3] McHargue, C. J. and Scott, J. L., Metall. Trans. A, 9A, 151 (1978).
- [4] Scace, R. I. and Slack, G. A., J. Chem. Phys. 30, 1551 (1959).
- [5] Knippenberg, W. F., Philips Research Reports, 18, 161 (1963).
- [6] Stull, D. R. and Prophet, H., JANAF Thermochemical Tables, Second edition, National Bureau of Standards Publication NSRDS-NBS 37 (June 1971).
- [7] Addamino, A., Speculations on the origins of the polytypism of SiC, in Silicon Carbide-1973, 179 (Univ. of S. Carolina Press, Columbia, S.C., 1973).
- [8] Ruff, O. and Korschak, M., Z. Electrochem. 32, 515 (1926).
- [9] Ruff, O., Trans. Electrochem. Soc. 68, 87 (1935).
- [10] Drowart, J., DeMaria, G., and Inghram, M. G., J. Chem. Phys. 29, 1015 (1958).
- [11] Weltner, W., Jr. and McLeod, D., Jr., J. Chem. Phys. 41, 235 (1964).
- [12] Drowart, J. and DeMaria, G., Thermodynamic study of the binary system carbon-silicon using a mass spectrometer, in Silicon Carbide. A High Temperature Semiconductor, J. R. O'Connor and J. Smiltens, eds., p. 16 (Pergamon Press, 1960).

- [13] Grieveson, P. and Alcock, C. B., The thermodynamics of metal silicides and silicon carbide, in Special Ceramics, P. Popper, ed., p. 183 (Heywood and Company Ltd., London, 1960).
- [14] Davis, S. G., Anthrop, D. F., and Searcy, A. W., J. Chem. Phys. 34, 659 (1961).
- [15] Mell, H. and Brodsky, M. H., Thin Film Solids, 46, 299 (1977).
- [16] Voronin, N. I., Makarova, N. L., and Yudin, B. F., Issled. v Obl. Khim. Silikatov i Okislov, Akad. Nauk SSSR, Sb. Statei, 203 (1965).
- [17] Ghoshtagore, R. N., Solid-State Electronics, 9, 178 (1966).
- [18] Paule, R. C. and Mandel, J., Analysis of Interlaboratory Measurements on the Vapor Pressure of Gold, National Bureau of Standards Special Publication 260-19 (January, 1970).
- [19] Cubicciotti, D., J. Phys. Chem. 70, 2410 (1966).
- [20] Behrens, R. G. and Rosenblatt, G. M., J. Chem. Thermodynamics, 4, 175 (1972).
- [21] Meschi, D., J. Phys. Chem. 76, 2947 (1972).
- [22] Roberts, J. Jr., and Searcy, A. W., High Temperature Science, 4, 411 (1972).
- [23] Greenberg, E., Natke, C. A., and Hubbard, W. N., J. Chem. Thermodynamics, 2, 193 (1970).
- [24] Searcy, A. W. and Beruto, D., J. Phys. Chem. 80, 425 (1976).
- [25] Searcy, A. W. and Beruto, D., J. Phys. Chem. 82, 163 (1978).

Discussion

Question (Weltner): You spoke of Drowart's data; did you restrict yourself when you were thinking of the molecular species and their properties to Drowart's discussion of them? There has been some recent work on SiC_2 by Verma and Nagaraj [Canadian J. Phys. 52, 1938 (1974)], and the vibrational frequencies have changed. Do you know of this?

Response (Rinehart): I am not aware of this.

Comment (Weltner): Well, the SiC_2 molecule is now pinned down quite thoroughly, I think, and very well done. The low vibrational frequencies have shifted considerably from earlier data. Now, I really didn't notice how good your agreement was between second and third law, but I just thought you would be interested.

Question (Hildenbrand): You mentioned the change in transmission efficiency with your mass filter, with mass discrimination effects between gold and silicon. But of course that would also affect the evaluation of these minor species partial pressures as well.

Is it possible to make reliable corrections for mass discrimination that are stable and would be useful over a wide mass range. Lots of people publish data with quadrupole instruments and never mention this problem.

(Editor's note): The response was not recorded but the main text contains some discussion of this point. Further pertinent information can be found elsewhere in this volume (see Bonnell and Hastie).

Moderator (Rosenblatt): Do you expect the mass signal to be independent of surface area from the geometry of the sampling system and beam collimation used in the mass spectrometer.

Response (Rinehart): The crystal was fairly large, and I don't think we were really sampling it all. We sampled only a small portion of that.